Light

GCSE Computer Science — Programming Testing and Robustness (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

When are syntax errors detected in a program?

- ○ **A.** During program execution
- ○ **B.** Before the program runs, during translation
- ○ **C.** Only when specific data is entered
- ○ **D.** After the program has finished running

Q2 (1 mark)

Which of the following would cause an infinite loop?

- A. `for i in range(10):`
- B. `while True:` with no break statement
- C. `if x > 5:`
- D. `print("Hello")`

Q3 (1 mark)

A program tries to open a file that does not exist. What type of error occurs?

- A. Syntax error
- B. Logic error
- C. Runtime error
- D. No error occurs

Q4 (1 mark)

Why is testing an important part of program development?

- A. To make the program run faster
- B. To reduce the size of the program file
- C. To add new features to the program
- D. To find and fix errors before the program is released

Q5 (1 mark)

What type of programming error is boundary testing specifically designed to detect?

- ○ **A.** Off-by-one errors in conditions
- ○ **B.** Syntax errors in variable names
- ○ **C.** Spelling mistakes in output messages
- ○ **D.** Missing import statements

Q6 (1 mark)

A program expects a numeric input for age. Which of the following is erroneous test data?

- ○ **A.** 25
- ○ **B.** 0
- ○ **C.** "twenty-five"
- ○ **D.** 100

Q7 (1 mark)

What does a length check validate?

- ○ **A.** That a number is within a specified range
- ○ **B.** That a field is not empty
- ○ **C.** That data matches a specific pattern

- ○ **D.** That data has the correct number of characters

Q8 (1 mark)

What is the purpose of a type check in validation?

- ○ **A.** To verify the data is the correct data type
- ○ **B.** To check that a file exists
- ○ **C.** To ensure passwords are encrypted
- ○ **D.** To count the number of records

Q9 (1 mark)

How does adding blank lines between sections of code improve a program?

- ○ **A.** It makes the program execute faster
- ○ **B.** It reduces the memory the program uses
- ○ **C.** It makes the code easier to read and understand
- ○ **D.** It prevents syntax errors from occurring

Q10 (1 mark)

Why should constant values be defined at the top of a program rather than written directly in the code?

- ○ **A.** It makes values easier to find and change

- ○ **B.** It makes the program run faster
- ○ **C.** It prevents all runtime errors
- ○ **D.** It reduces the file size

Q11 (1 mark)

Which is the most common method of authentication?

- ○ **A.** Entering a CAPTCHA
- ○ **B.** Clicking a checkbox
- ○ **C.** Accepting terms and conditions
- ○ **D.** Username and password combination

Q12 (1 mark)

How can print statements help during debugging?

- ○ **A.** They show the values of variables at specific points
- ○ **B.** They automatically fix errors in the code
- ○ **C.** They make the program run without errors
- ○ **D.** They remove the need for testing

Q13 (2 marks)

Consider the following Python code:

```
result = max(10, 20, 30
print(result)
```

What type of error does this code contain?

- ○ **A.** Logic error
- ○ **B.** Runtime error
- ○ **C.** Syntax error
- ○ **D.** No error

Q14 (2 marks)

A program should display "Pass" if a score is 50 or above. Consider the code:

```
score = 50
if score > 50:
    print("Pass")
else:
    print("Fail")
```

The program outputs "Fail" for a score of 50. What type of error is this?

- ○ **A.** Syntax error
- ○ **B.** Runtime error
- ○ **C.** Logic error
- ○ **D.** Type error

Q15 (2 marks)

Which code correctly implements a presence check to ensure a username is not empty?

- A. `if username == "":`
- B. `if username > 0:`
- C. `if username = "":`
- D. `if len(username) > 0:`

Q16 (2 marks)

Which code correctly checks if a string contains only digits before converting to an integer?

- A. `if user_input.isdigit():`
- B. `if user_input == int:`
- C. `if type(user_input) == int:`
- D. `if user_input.isnumber():`

Q17 (2 marks)

A program accepts ages between 18 and 65. Which row correctly categorises ALL the test data?

- A. Normal: 17 | Boundary: 18, 65 | Erroneous: -5
- B. Normal: 40 | Boundary: 18, 65 | Erroneous: "forty"
- C. Normal: 18, 65 | Boundary: 40 | Erroneous: 100

- ○ **D.** Normal: -5 | Boundary: 17, 66 | Erroneous: 40

Q18 (2 marks)

What is input sanitisation?

- ○ **A.** Deleting all user input immediately
- ○ **B.** Encrypting all input data
- ○ **C.** Displaying input in uppercase only
- ○ **D.** Removing or escaping potentially harmful characters from input

Q19 (2 marks)

Consider the following code:

```
value = "12.5"
number = int(value)
print(number)
```

What type of error will occur?

- ○ **A.** Runtime error — cannot convert decimal string to int
- ○ **B.** No error — outputs 12
- ○ **C.** Syntax error — incorrect function usage
- ○ **D.** Logic error — outputs wrong value

Q20 (2 marks)

Why does breaking code into functions and procedures improve maintainability?

- ○ **A.** It reduces repetition and makes code easier to update
- ○ **B.** It makes the program execute faster
- ○ **C.** It prevents all types of errors
- ○ **D.** It automatically adds comments to the code

Q21 (2 marks)

A date must be entered in the format DD/MM/YYYY. What type of validation check is needed?

- ○ **A.** Range check
- ○ **B.** Presence check
- ○ **C.** Length check
- ○ **D.** Format check

Q22 (2 marks)

Why should a system lock an account after several failed login attempts?

- ○ **A.** To save memory on the server
- ○ **B.** To encourage users to create shorter passwords
- ○ **C.** To reduce network traffic

- ○ **D.** To prevent brute force password attacks

Q23 (2 marks)

What is iterative testing?

- ○ **A.** Testing each part of the program as it is developed
- ○ **B.** Testing the program only once at the end
- ○ **C.** Having users test the program
- ○ **D.** Removing all bugs automatically

Q24 (3 marks)

A program accepts a month number (1-12). A complete test plan should include which test values?

- ○ **A.** Just 1, 6, and 12
- ○ **B.** Only positive numbers
- ○ **C.** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
- ○ **D.** 0, 1, 6, 12, 13, and "June"

Q25 (3 marks)

The following code should count from 1 to 10 inclusive, but it only outputs 1 to 9:

```
count = 1
while count < 10:
```

```
    print(count)
    count = count + 1
```

What is the error and how should it be fixed?

- ○ **A.** Syntax error: change 'while' to 'for'
- ○ **B.** Logic error: change 'count < 10' to 'count <= 10'
- ○ **C.** Runtime error: change 'count + 1' to 'count + 2'
- ○ **D.** Logic error: change 'count = 1' to 'count = 0'

Q26 (3 marks)

Consider this code for validating a quantity between 1 and 99:

```
valid = False
while not valid:
    qty = input("Enter quantity (1-99): ")
    if qty.isdigit():
        qty = int(qty)
        if qty >= 1 and qty <= 99:
            valid = True
        else:
            print("Must be 1-99")
    else:
        print("Enter a number")
```

What validation does this code perform?

- ○ **A.** Only range check
- ○ **B.** Only type check
- ○ **C.** Type check then range check

- ○ **D.** Range check then type check

Q27 (3 marks)

Consider the following code:

```
total = 0
for i in range(1, 4):
    total = total + i * 2
print(total)
```

What value is output?

- ○ **A.** 6
- ○ **B.** 12
- ○ **C.** 20
- ○ **D.** 8

Q28 (3 marks)

For a robust username field, which combination of validation checks should be applied?

- ○ **A.** Presence check only
- ○ **B.** Length check only
- ○ **C.** Presence check, length check, and pattern check
- ○ **D.** Range check and type check

Q29 (3 marks)

Match the testing type to its description:
1. Alpha testing
2. Beta testing
3. Acceptance testing
A. Testing by the client to verify requirements are met
B. Testing by developers within the company
C. Testing by a limited group of external users

- ○ **A.** 1-A, 2-B, 3-C
- ○ **B.** 1-B, 2-C, 3-A
- ○ **C.** 1-C, 2-A, 3-B
- ○ **D.** 1-B, 2-A, 3-C

Q30 (3 marks)

Which error message demonstrates good practice for a robust program?

- ○ **A.** "Error"
- ○ **B.** "ValueError: invalid literal for int() with base 10"
- ○ **C.** "Please enter a number between 1 and 100"
- ○ **D.** "Error code: 0x8007000E"

Total: **55** marks • Time: **30 min**

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